MRSpinDynamics

Python User Manual

Python magnetic-resonance spin-dynamics workspace

Simulation workflows for magnetic-resonance spin dynamics in possibly non-uniform and time-varying fields, with validated MATLAB-compatible spin-1/2 NMR Bloch kernels, Python-native analysis tools, scoped scalar-coupled spin-1/2 models, and pulsed NQR extensions.

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Contents

1 Physical Scope and Assumptions

The MATLAB-compatible workflow layer in PythonSpinDynamics models a bath of uncoupled spin-1/2 nuclei evolving in a static, spatially non-uniform, or time-varying main field $B_0(\mathbf{r}, t)$ and applied RF fields. The elementary object in that layer is an isochromat: a subensemble of spin-1/2 nuclei that shares an offset, RF sensitivity, relaxation constants, and probe response.

Modeled directly in the core Bloch workflows. Independent spin-1/2 magnetization vectors; offset distributions from $B_0(\mathbf{r}, t)$; RF rotations; free precession; phenomenological T_1 and T_2 relaxation; probe transmit and receive filtering; optional deterministic radiation damping; optional motion through field maps.

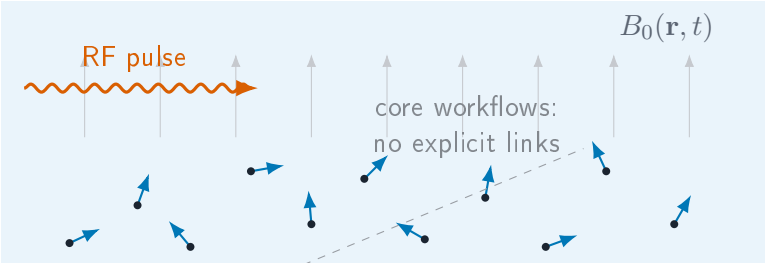
Scalar-coupling extension. The `spin_dynamics.coupling` namespace adds small dense spin-1/2 systems with scalar J couplings, analytic low-field J-editing models, ideal TANGO-B filter curves, and initial SLIC simulations. These models are intentionally scoped and are documented in Chapter 5.

Pulsed NQR extension. The `spin_dynamics.nqr` namespace adds dense single-site quadrupolar spin tools for selective pulsed NQR, including spin-1 and spin-3/2 transition metadata, single-crystal and powder orientation averages, SLSE echo trains, two-frequency population transfer, EFG inhomogeneity, finite-window SLSE spectra, and weak-B0 perturbations. These models are documented in Chapter 6.

Still outside the package scope. Dipolar coupling networks, chemical exchange, arbitrary nonselective multi-quantum pulse sequences, shaped sample microstructure beyond the supplied field and density maps, microscopic spin noise, and large sparse coupled-spin simulations. Spin-spin effects enter the Bloch workflows only indirectly when represented by effective relaxation times such as T_2 or T_2^*.

This scope is deliberate. The main package follows MATLAB's classical Bloch/isochromat viewpoint, which is appropriate for many pulse, probe, imaging, and relaxation studies where the spins can be treated as independent subensembles. The coupled-spin namespace adds targeted quantum spin-1/2 tools for scalar J-coupling phenomena without turning the full package into a general quantum spin Hamiltonian framework.

1.1 Spin Bath Cartoon



Independent isochromats sample the local field and RF sensitivity.

1.2 Consequences for Users

- Use offset grids, field maps, or moving particles to represent spatially varying B_0, transmit B_1, and receive B_1 effects.
- Use relaxation constants to represent ensemble dephasing and energy exchange that are outside the explicit isochromat model.
- Use `spin_dynamics.coupling` when a small spin-1/2 scalar J-coupled model is the phenomenon of interest.
- Use `spin_dynamics.nqr` when selective pulsed NQR of a small quadrupolar site is the phenomenon of interest.
- Do not use the package for chemical exchange, large coupled-spin networks, or arbitrary nonselective multi-quantum coherence pathways.

2 Overview

PythonSpinDynamics is a Python port of the MATLAB spin-dynamics simulation package. The MATLAB tree under `../MATLABSpinDynamics/SpinDynamicsUpdated/Version_2/code` remains the reference for numerical behavior, while this workspace provides a typed, testable Python API for simulations, validation fixtures, examples, and new analysis workflows.

The supported surface currently includes ideal, tuned, untuned, and matched probe CPMG workflows; finite echo trains; FID; phase-encoded imaging; diffusion CPMG; time-varying-field CPMG; WURST inversion and matched WURST-CPMG; radiation damping; moving-isochromat motion helpers; received-signal noise; inverse-Laplace analysis; compact pulse-optimization scaffolds; and small spin-1/2 scalar-coupling tools for low-field J-editing, TANGO-B filtering, and SLIC-style spin-lock response; and early pulsed NQR helpers for quadrupolar sites, powder averaging, SLSE, two-frequency population transfer, EFG inhomogeneity, spin-3/2 transition metadata, and weak-static-field spectra.

2.1 Package Layout

Path	Purpose
`src/spin_dynamics/`	Installable Python package.
`examples/`	CLI examples and plotting scripts.
`tests/`	Smoke tests, validation tests, and unit tests.
`validation/fixtures/`	MATLAB/Octave fixture arrays.
`validation/octave/`	Fixture generation scripts.
`docs/python_api/`	Lightweight Markdown API notes.
`docs/validation_results.md`	Current validation status and tolerance notes.
`docs/radiation_damping.md`	Focused radiation-damping model notes.

3 Installation and Test Runs

3.1 Editable Install

PythonSpinDynamics requires Python 3.10 or newer. The core install depends on NumPy 1.24 or newer and does not require MATLAB at runtime. MATLAB is only needed to regenerate the complete MATLAB reference fixture set; Octave can regenerate the smaller dependency-light fixtures.

From `PythonSpinDynamics`, install the package into an active environment:

```
python -m pip install -e .
```

Optional extras are provided for development, plotting, benchmarking, and SciPy-backed optimization:

```
python -m pip install -e ".[dev,opt,plot,bench]"
```

The core package depends only on NumPy. The `opt` extra enables SciPy-backed continuous optimizers, non-negative inverse-Laplace solves, and MATLAB `.mat` result import/export helpers. The `plot` extra enables the Matplotlib examples.

3.2 Tests

Use the smoke tier during normal editing:

```
python -m unittest tests.smoke_tests
```

Run the full validation suite before committing numerical or workflow changes:

```
python -m unittest discover -s tests
```

4 Model Conventions

4.1 Coherence Ordering

Low-level kernels preserve MATLAB's coherence ordering:

$$\mathbf{M} = \begin{bmatrix} M_0 & M_- & M_+ \end{bmatrix}^{\mathsf{T}}.$$

This ordering differs from many Bloch-vector presentations, so new kernels and validation tests should state the expected ordering explicitly.

4.2 Normalized Time

The core MATLAB routines usually normalize time to the nominal RF amplitude $\omega_1 = 1$. In these units, a 90-degree hard pulse has duration

$$t_{90} = \frac{\pi}{2},$$

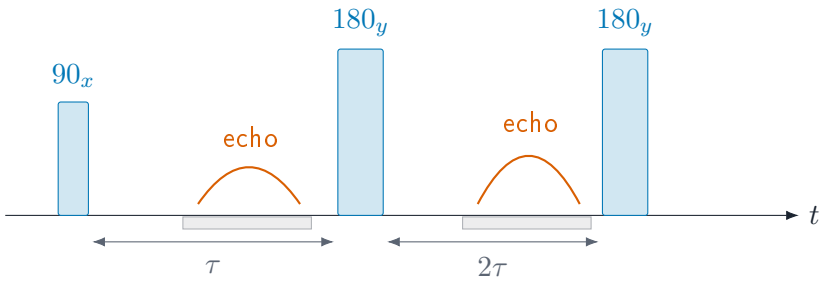
and a 180-degree hard pulse has duration

$$t_{180} = \pi.$$

Workflow helpers that expose physical seconds convert into these normalized units before calling the lower-level kernels.

4.3 Pulse Timing Cartoon

A minimal spin-echo or CPMG building block is a sequence of RF pulses, free precession windows, and acquisition windows. The package represents these as ordered intervals with pulse matrices and free-precession delays.



4.4 Effective Rotations

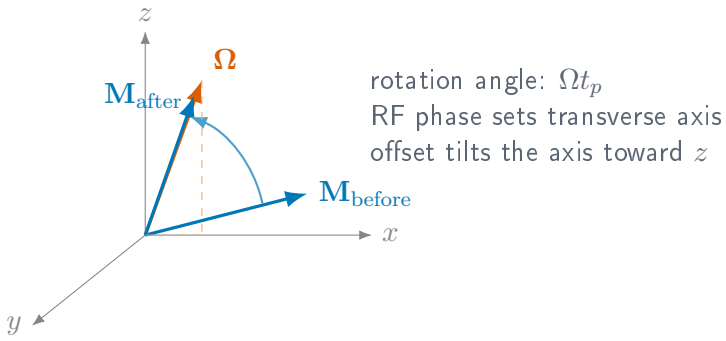
For an isochromat with normalized offset $\Delta\omega$, RF amplitude ω_1, phase ϕ, and pulse duration t_p, the rotating-frame effective field is

$$\boldsymbol{\Omega} = \begin{bmatrix} \omega_1 \cos\phi \\ \omega_1 \sin\phi \\ \Delta\omega \end{bmatrix}, \qquad \Omega = \sqrt{\omega_1^2 + \Delta\omega^2}.$$

The magnetization update is a rotation by angle Ωt_p about $\boldsymbol{\Omega}/\Omega$. The Python rotation helpers expose this calculation through `spin_dynamics.core.rotations` and return array shapes chosen to match the MATLAB reference fixtures.

4.5 Rotation Cartoon

In the rotating frame, a rectangular RF segment rotates the magnetization about an effective axis set by RF phase, RF amplitude, and off-resonance offset.



4.6 Relaxation

Free-precession intervals with relaxation use the usual exponential factors:

$$M_{xy}(t+\Delta t) = M_{xy}(t)\exp\left(-\frac{\Delta t}{T_2}\right)\exp(-\mathrm{i}\Delta\omega\,\Delta t),$$

$$M_z(t+\Delta t) = M_{\mathrm{th}} + (M_z(t) - M_{\mathrm{th}})\exp\left(-\frac{\Delta t}{T_1}\right).$$

Finite acquisition workflows assemble these intervals from pulse-sequence parameters and then call the `arb1D`-style kernels.

4.7 Echo Conversion

The echo utilities direct-sum a received spectrum $S(\Delta\omega)$ over the offset grid:

$$e(t) = \sum_k S(\Delta\omega_k)\exp(\mathrm{i}\Delta\omega_k t)\,\Delta\omega.$$

The implementation uses the same direct-sum convention as the MATLAB `calc_time_domain_echo` reference path. FID traces use the corresponding FID conversion helper.

4.8 Radiation Damping

The deterministic radiation-damping back-action model follows the local Measurements text convention:

$$T_{\mathrm{rd}} = \frac{2}{\gamma\mu_0\eta M_0 Q},$$

where γ is the gyromagnetic ratio, η is the magnetic-energy fill factor, M_0 is the equilibrium magnetization density, and Q is the loaded probe quality factor.

The analytic on-resonance hard-pulse FID envelope used by the test suite is

$$|M_{xy}(t)| = \frac{M_0}{\cosh\left(t/T_{\mathrm{rd}} - \log\tan(\theta/2)\right)}.$$

The finite-sequence implementation can add feedback during free-precession and acquisition windows, and can optionally apply an operator-split approximation during RF pulse matrices.

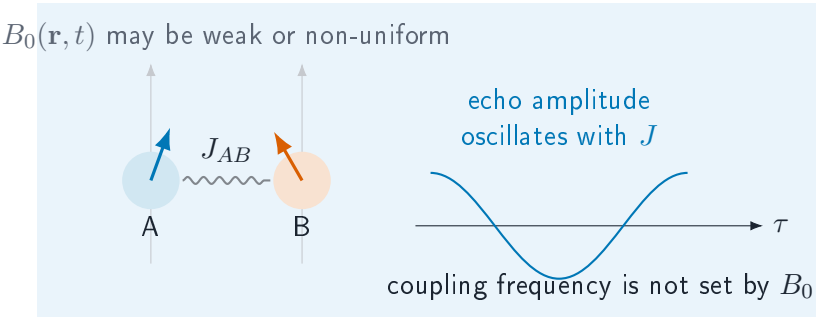
5 Scalar J-Coupling Models

Scalar J coupling is field independent, so it can remain visible in weak or grossly inhomogeneous magnetic fields where chemical-shift resolution is compressed. The `spin_dynamics.coupling` namespace adds a scoped set of spin-1/2 tools for these cases. It is separate from the MATLAB-compatible Bloch workflow layer: Bloch workflows propagate independent isochromats, while the coupling layer propagates small dense Hilbert-space systems or evaluates closed-form J-editing response curves.

Use this layer for: heteronuclear low-field J-editing curves, known-J spectrum fitting, ideal TANGO-B filter selectivity, and exploratory two-spin or few-spin scalar-coupled Hamiltonian simulations.

Current limits: spin-1/2 only; dense matrices only; no relaxation superoperators; no chemical exchange; no quadrupolar nuclei; no large sparse coupled-spin networks; no full pulse-sequence compiler for arbitrary multi-quantum pathways.

5.1 J-Coupled Spin Cartoon



5.2 Dense Spin-1/2 Hamiltonians

A coupled system is built from offsets ν_i in hertz and symmetric scalar couplings J_{ij} in hertz:

```
from spin_dynamics.coupling import (
    coupled_spin_system,
    isotropic_j_hamiltonian,
    zeeman_hamiltonian,
)

system = coupled_spin_system(
    offsets_hz=[-0.35, 0.35],
    couplings_hz=[[0.0, 7.0], [7.0, 0.0]],
    labels=["A", "B"],
)
hamiltonian = zeeman_hamiltonian(system) + isotropic_j_hamiltonian(system)
```

Hamiltonian builders return angular-frequency matrices in radians per second and use spin-1/2 operators

with eigenvalues $\pm 1/2$. The rotating-frame offset Hamiltonian is

$$H_Z = 2\pi \sum_i \nu_i I_{iz}.$$

For weakly coupled systems, the secular scalar Hamiltonian is

$$H_J^{\mathrm{sec}} = 2\pi \sum_{i<j} J_{ij} I_{iz} I_{jz},$$

while the isotropic scalar Hamiltonian is

$$H_J^{\mathrm{iso}} = 2\pi \sum_{i<j} J_{ij} \left(I_{ix} I_{jx} + I_{iy} I_{jy} + I_{iz} I_{jz} \right).$$

The dense propagator evaluates

$$\rho(t) = U(t)\rho(0)U^{\dagger}(t), \qquad U(t) = \exp\left(-\mathrm{i}Ht\right),$$

and is intended for small systems where dense matrices are acceptable.

5.3 Inhomogeneous B0/B1 Isochromats

The bridge between the coupled-spin layer and the Bloch workflow layer is a coupled isochromat ensemble. Each isochromat contains the same scalar-coupled spin network, but samples a different local field and receive weight. For isochromat k, the local spin offsets are

$$\nu_{i,k} = \nu_i^{(0)} + s_i \, \Delta\nu_k,$$

where $\nu_i^{(0)}$ is the base spin offset, $\Delta\nu_k$ is the local B0 offset in hertz, and s_i is a per-spin offset scale. The default $s_i = 1$ adds the same field offset to every spin. Heteronuclear or carrier-specific models can supply different scales.

During an RF or spin-lock interval, the nominal nutation frequency is scaled by the local transmit field:

$$\omega_{1,k} = b_{1,k}^{\mathrm{tx}} \omega_1.$$

The ensemble signal is accumulated as

$$S = \sum_k w_k b_{1,k}^{\mathrm{rx}} \, \mathrm{Tr}(\rho_k D),$$

where w_k is the isochromat weight, $b_{1,k}^{\mathrm{rx}}$ is the local receive scale, and D is the detection operator.

```
from spin_dynamics.coupling import (
    coupled_isochromat_ensemble,
    free_precession_step,
    rf_step,
    simulate_coupled_isochromat_sequence,
)

ensemble = coupled_isochromat_ensemble(
    system,
    b0_offsets_hz=[-2.0, 0.0, 2.0],
```

```
    weights=[0.25, 0.5, 0.25],
    b1_tx_scale=[0.9, 1.0, 1.1],
    b1_rx_scale=[1.0, 1.0, 0.8],
)

result = simulate_coupled_isochromat_sequence(
    ensemble,
    [
        rf_step(duration=0.25, nutation_hz=1.0, phase=1.5708),
        free_precession_step(duration=0.01),
    ],
    initial_axis="x",
    detect_axis="x",
)
```

For time-varying fields, each sequence step may override the ensemble B0 offsets or B1 transmit scale. This keeps the first implementation close to the existing isochromat approach: the coupled-spin Hilbert space is dense and small, while field inhomogeneity is represented by an outer ensemble sum.

5.4 Low-Field J-Editing

Analytic J-editing helpers model response curves as sums of field-independent cosine modulations:

$$s(\tau) = b + \sum_m a_m \cos^{p_m}(2\pi N J_m \tau)\,,$$

where τ is the encoding time, N is the number of encoding cycles, J_m is a known or hypothesized coupling in hertz, a_m is its amplitude, and p_m is an optional multiplicity exponent. Proton-detected models use $p_m = 1$. Carbon-detected CH_n groups use $p_m = n$.

```
from spin_dynamics.coupling import (
    carbon_detected_j_modulation,
    fit_known_j_spectrum,
    j_modulation_curve,
)

signal = j_modulation_curve(
    encoding_times,
    couplings_hz=[125.0, 160.0],
    amplitudes=[0.85, 0.15],
)

carbon_signal = carbon_detected_j_modulation(
    encoding_times,
    couplings_hz=[125.0, 160.0],
    abundances=[0.85, 0.15],
    proton_counts=[2, 1],
)

fit = fit_known_j_spectrum(
    encoding_times,
    signal,
    couplings_hz=[125.0, 160.0],
    include_background=False,
```

```
)
```

`fit_known_j_spectrum` solves a linear least-squares problem once the candidate J_m values are supplied. It returns amplitudes, optional background, fitted signal, residual, rank, and residual norm.

5.5 TANGO-B Filtering

The ideal TANGO-B helper evaluates the coupled-spin transverse filter envelope

$$A(J; t_d) = \sin^2(\pi J t_d)\,.$$

When a target coupling $J_\star$ and positive odd order n are supplied, the delay is selected as

$$t_d = \frac{n}{2J_\star}.$$

```
from spin_dynamics.coupling import tango_b_filter

response = tango_b_filter(
    couplings_hz,
    target_coupling_hz=160.0,
    order=3,
)
```

This model is idealized: it reports the scalar-coupling selectivity envelope, not RF imperfections, relaxation, diffusion, or probe filtering.

5.6 SLIC Response

Spin-lock induced crossing (SLIC) is modeled by adding an RF spin-lock Hamiltonian to the static coupled-spin Hamiltonian:

$$H_{\mathrm{SLIC}} = H_Z + H_J^{\mathrm{iso}} + 2\pi\omega_1 \sum_i I_{ix}.$$

The helper scans spin-lock nutation frequencies ω_1 in hertz and returns the remaining transverse magnetization and its corresponding dip:

```
from spin_dynamics.coupling import (
    coupled_spin_system,
    simulate_slic_spectrum,
    two_spin_slic_transfer_time,
)

system = coupled_spin_system(
    offsets_hz=[-0.35, 0.35],
    couplings_hz=[[0.0, 7.0], [7.0, 0.0]],
)
duration = two_spin_slic_transfer_time(offset_difference_hz=0.7)
result = simulate_slic_spectrum(
    system,
    nutation_frequencies_hz,
    spin_lock_time=duration,
)
```

For a nearly equivalent two-spin system, the SLIC dip is expected near the coupling frequency. The helper is exploratory and dense; broad homonuclear networks will need sparse or specialized methods.

6 Pulsed NQR Models

Nuclear quadrupole resonance is driven by the interaction between a nuclear quadrupole moment and the local electric-field-gradient (EFG) tensor. The `spin_dynamics.nqr` namespace adds a scoped quadrupolar extension for solid-state pulsed NQR experiments. It is intentionally separate from the spin-1/2 Bloch and scalar-coupling layers.

Use this layer for: selective pulsed NQR of one spin-1 quadrupolar site, spin-1 and spin-3/2 transition metadata, fixed-orientation single-crystal calculations, powder averages over local EFG orientations, SLSE echo trains, two-frequency population transfer, static EFG inhomogeneity, finite-window SLSE spectra, and weak-B0 line splitting.

Current limits: dense single-site matrices only; selective RF pulses only; spin-3/2 pulsed dynamics pending a degenerate-doublet RF manifold model; phenomenological relaxation rather than microscopic Redfield/dipolar rates; no simultaneous multi-frequency excitation; no full nonselective shaped-pulse Hamiltonian propagation; no multi-site coupling between quadrupolar nuclei.

6.1 Quadrupolar Site Model

A quadrupolar site is described in the EFG principal-axis system. The zero-field Hamiltonian used by the Python helper is

$$H_Q = 2\pi\nu_{\mathrm{scale}}\left[3I_z^2 - I(I+1)\mathbf{1} + \eta\left(I_x^2 - I_y^2\right)\right],$$

where I is the spin quantum number, ν_Q is the quadrupole-frequency parameter in hertz, and $0 \le \eta \le 1$ is the EFG asymmetry parameter. Hamiltonians are represented in radians per second. For spin-1, the package uses $\nu_{\mathrm{scale}} = \nu_Q/3$, matching the nitrogen-14 convention where the zero-field lines are $\nu_x = \nu_Q(1+\eta/3)$, $\nu_y = \nu_Q(1-\eta/3)$, and $\nu_z = 2\eta\nu_Q/3$. For spin-3/2, $\nu_{\mathrm{scale}} = \nu_Q/6$, so the zero-field chlorine-style NQR line is

$$\nu_{\mathrm{NQR}} = \nu_Q\sqrt{1+\eta^2/3}.$$

Zero-frequency Kramers-doublet transitions are omitted from the transition inventory.

Weak Zeeman perturbations can be added as

$$H_Z = -2\pi\gamma\,\mathbf{B}_0\cdot\mathbf{I},$$

where $\mathbf{B}_0$ is expressed in the EFG principal-axis system for the current crystallite orientation. The default is $B_0 = 0$, which is the usual NQR case.

```
from spin_dynamics.nqr import QuadrupolarSite, diagonalize_site

site = QuadrupolarSite(
    spin=1,
    isotope="14N",
    quadrupole_frequency_hz=900e3,
    eta=0.3,
)

eigensystem = diagonalize_site(site)
for transition in eigensystem.transitions:
    print(transition.label, transition.frequency_hz)
```

For spin-1 at zero field, transitions are labeled by their dominant RF polarization in the EFG principal-axis system: `x`, `y`, and `z`. For the example above the line positions are approximately 990, 810, and 180 kHz.

Spin-3/2 transition metadata can be inspected with the same diagonalization helper:

```
chlorine = QuadrupolarSite(
    spin=1.5,
    isotope="35Cl",
    quadrupole_frequency_hz=30e6,
    eta=0.1,
    gamma_hz_per_t=4.171e6,
)

for transition in diagonalize_site(chlorine).transitions:
    print(transition.frequency_hz)
```

6.2 Selective Pulse Approximation

Most practical pulsed NQR RF pulses are narrowband relative to the separation between quadrupolar transitions. The first NQR implementation uses this selective limit: one pulse addresses one transition and acts as an embedded two-level rotation inside the full $(2I+1)$-level Hilbert space.

For a transition $|a\rangle \leftrightarrow |b\rangle$, the transition dipole vector is

$$\mathbf{d}_{ab} = \left(\langle a|I_x|b\rangle, \langle a|I_y|b\rangle, \langle a|I_z|b\rangle\right).$$

For a local RF direction $\hat{\mathbf{b}}_1$, the complex drive projection is proportional to $\hat{\mathbf{b}}_1 \cdot \mathbf{d}_{ab}$. Preserving this signed complex projection is essential for powder averages because transmit and receive phases must pair consistently across crystallites.

```
from spin_dynamics.nqr import SelectivePulse, apply_selective_pulse

pulse = SelectivePulse(
    "x",
    duration_seconds=50e-6,
    nutation_hz=10e3,
)
```

The pulse helper also supports an optional RF frequency. If omitted, the RF is placed on the selected transition. If supplied, the difference between the RF frequency and transition frequency is treated as a two-level detuning.

The selective-pulse propagation routines currently support spin-1 sites only. Spin-3/2 nuclei such as $^{35}\mathrm{Cl}$ and $^{37}\mathrm{Cl}$ have degenerate doublets at zero field; their pulsed response needs an RF model that acts on the full doublet manifold rather than an embedded two-level transition. Static spin-3/2 transition frequencies and weak-B0 spectra are available now, but spin-3/2 SLSE/FID pulse simulations deliberately raise a clear error.

6.3 Orientations and Powder Averages

NQR is a solid-state experiment, so the signal depends on the orientation of the RF field relative to the local EFG principal axes. A single crystal is one fixed orientation. A powder is represented by a normalized spherical quadrature grid:

```
from spin_dynamics.nqr import powder_average_grid, single_crystal_orientation

single = single_crystal_orientation(alpha=0.0, beta=1.57079632679)
powder = powder_average_grid(n_theta=16, n_phi=32)
```

For weak-B0 calculations, each orientation may also specify the direction of B_0 in the EFG frame. If no separate B_0 direction is supplied, the current helper uses the local RF direction as the default field direction for that sample.

Weak-field spectra use a correlated powder grid for the static field and RF field directions. The helper `b0_b1_powder_average_grid` samples the orientation of B_0 in the EFG frame and the rotation angle of B_1 around B_0, preserving a chosen lab-frame angle between the two fields. For a conventional transverse coil one uses $B_1 \perp B_0$.

6.4 Weak Static B0 Spectra

Weak static magnetic fields are modeled by exact diagonalization of

$$H = H_Q + H_Z, \qquad H_Z = -2\pi\gamma\,\mathbf{B}_0 \cdot \mathbf{I}.$$

The helper is intended for the NQR regime

$$\epsilon_Z = \frac{|\gamma B_0|}{\nu_{\mathrm{ref}}} \ll 1,$$

and reports the largest perturbation ratio in the result. It can warn or raise if the requested field is no longer weak compared with the selected NQR line.

```
from spin_dynamics.nqr import simulate_weak_b0_spectrum

weak = simulate_weak_b0_spectrum(
    chlorine,
    b0_tesla=1e-3,
    orientations="powder",
    broadening_hz=200.0,
    weak_ratio_threshold=0.05,
)

print(weak.max_perturbation_ratio)
```

For each crystallite, the code diagonalizes $H_Q + H_Z$, keeps transitions near the selected zero-field NQR line, and weights them by the RF projection

$$I_{ab} \propto w_k \left|\hat{\mathbf{b}}_{1,k} \cdot \mathbf{d}_{ab,k}\right|^2,$$

where w_k is the orientation weight. This RF projection is important for spin-3/2 in weak fields: the eigenvalue splitting can contain several nearby transitions, but transitions dark to the coil should not contribute to the plotted spectrum. In the axial limit $\eta = 0$, $B_0 \parallel z_{\mathrm{PAS}}$, and $B_1 \perp B_0$, a spin-3/2 site gives the expected pair $\nu_Q \pm \gamma B_0$.

6.5 SLSE Detection

The spin-lock spin-echo (SLSE) sequence is the classic pulsed NQR detection sequence. The helper models it as repeated selective pulses on a spin-1 detection transition and reports echo amplitudes at the requested echo spacing. Echo decay can be represented by a scalar T_{2e} envelope or by a phenomenological Liouville-space relaxation model with T_1 and T_2.

```
from spin_dynamics.nqr import simulate_slse, slse_sequence

sequence = slse_sequence(
    "x",
    pulse_duration_seconds=25e-6,
    nutation_hz=10e3,
    echo_spacing_seconds=1e-3,
    num_echoes=16,
)

result = simulate_slse(
    site,
    sequence,
    orientations="powder",
    t2e_seconds=20e-3,
)
```

The returned `SLSEResult` includes echo times, powder-averaged echo amplitudes, per-orientation echo amplitudes, orientation weights, transition metadata, and the eigensystem used for the first orientation. When a relaxation model is supplied, the result also reports local cycle eigenvalues and cycle-derived effective decay times.

```
from spin_dynamics.nqr import NQRRelaxationModel

relaxed = simulate_slse(
    site,
    sequence,
    orientations="powder",
    relaxation=NQRRelaxationModel(t1_seconds=1.0, t2_seconds=20e-3),
)
```

Offset and pulse-period sweeps are available as compact diagnostics for the SLSE modulation:

```
from spin_dynamics.nqr import simulate_slse_offset_sweep

sweep = simulate_slse_offset_sweep(
    site,
    "x",
    offsets_hz=[-2e3, 0.0, 2e3],
    pulse_duration_seconds=25e-6,
    nutation_hz=10e3,
    echo_spacing_seconds=500e-6,
    relaxation=NQRRelaxationModel(t2_seconds=20e-3),
)
```

6.6 EFG Inhomogeneity and SLSE Spectra

Static EFG disorder is modeled as an isochromat ensemble of independent quadrupolar sites. Each isochromat has its own ν_Q, η, transition frequency, and normalized weight. This covers impurity-induced broadening, static temperature gradients, and other inhomogeneous mechanisms analogous to NMR isochromat broadening.

```
from spin_dynamics.nqr import (
    SelectivePulse,
    gaussian_efg_distribution,
    simulate_fid_efg_distribution,
)
import numpy as np

distribution = gaussian_efg_distribution(
    site,
    quadrupole_std_hz=2e3,
    samples=41,
)

fid = simulate_fid_efg_distribution(
    distribution,
    "x",
    times_seconds=np.linspace(0.0, 20e-3, 512),
    excitation=SelectivePulse("x", duration_seconds=2.5e-6, nutation_hz=100e3),
)
```

The distribution simulators check whether a coarse EFG grid can artificially rephase within the simulated duration. Increase the number of isochromats when the warning appears, or pass `rephase_action="ignore"` only after a convergence check.

In an SLSE experiment, the spectrum is not the FFT of the echo train. It is the FFT of the averaged echo acquired over a receiver window T_{acq} centered on a selected echo, with T_{acq} shorter than the pulse spacing. The finite acquisition window itself broadens the spectrum:

```
from spin_dynamics.nqr import simulate_slse_acquisition_spectrum

slse_spectrum = simulate_slse_acquisition_spectrum(
    distribution,
    sequence,
    acquisition_duration_seconds=200e-6,
    acquisition_points=256,
    echo_index=-1,
    noise={"target_snr": 20.0, "seed": 123},
    deconvolution_strength=1e-2,
)
```

Noise is added to the complex time-domain acquisition before the FFT. The optional deconvolution performs a regularized inversion of the same finite-window FFT operator. It is useful for exploring acquisition-window broadening, but it can amplify noise and should be checked across plausible SNR values.

6.7 Two-Frequency Population Transfer

The two-frequency NQR workflow follows the perturbation-plus-detection idea used in 2D NQR. A selective pulse on one transition changes populations in the shared three-level spin-1 manifold; a later SLSE block detects the changed amplitude on another transition. The normalized diagnostic is

$$\Delta(p, q) = \frac{S(p, q)}{S_0(q)} - 1,$$

where p is the perturbation transition and q is the detection transition.

```
from spin_dynamics.nqr import SelectivePulse, simulate_population_transfer

transfer = simulate_population_transfer(
    site,
    SelectivePulse("x", duration_seconds=50e-6, nutation_hz=10e3),
    sequence,
    orientations="powder",
)

print(transfer.normalized_difference)
```

The diagnostic plotting example constructs a compact 3×3 map over the spin-1 x, y, and z transitions. Off-diagonal features indicate that perturbation and detection frequencies are connected through the same quadrupolar site.

7 First Workflows

7.1 CPMG

Application code should normally start with the public workflow runners:

```
from spin_dynamics.workflows import run_tuned_cpmg

result = run_tuned_cpmg(numpts=101, maxoffs=10)
print(result.del_w.shape)
print(result.echo.shape)
print(result.snr)
```

The asymptotic CPMG runners return `CPMGResult` objects with the offset grid, asymptotic magnetization, received spectrum, time-domain echo, acquisition time vector, optional SNR, and probe label.

7.2 Finite Echo Trains

```
from spin_dynamics.workflows import run_matched_cpmg_train

train = run_matched_cpmg_train(
    numpts=51,
    num_echoes=8,
    auto_refine_grid=True,
    num_workers=None,
)
```

Finite train results contain one acquired spectrum and one echo per echo index. They also include trapezoidal echo integrals and physical echo-center times. For long trains, use the rephasing checks or enable `auto_refine_grid=True` so the offset grid is refined before RF matrices are built.

7.2.1 CPMG-IR Trains

Finite inversion-recovery trains add an inversion delay axis before the CPMG readout:

```
from spin_dynamics.workflows import run_matched_cpmg_ir_train

ir = run_matched_cpmg_ir_train(
    numpts=31,
    num_echoes=4,
    tauvect=[0.5e-3, 2e-3, 6e-3, 12e-3],
    auto_refine_grid=True,
)
```

The returned arrays are indexed by inversion delay first and echo number second. This makes the workflow useful for compact T_1-encoded echo-train tests without manually assembling the inversion pulse, delay, excitation, and refocusing blocks.

7.2.2 Probe Parameter Sweeps

Probe-aware CPMG workflows can also be swept over resonator Q or mistuning:

```
from spin_dynamics.workflows import run_tuned_q_sweep

sweep = run_tuned_q_sweep(q_values=[20, 50, 100], numpts=51)
```

Use the finite-train sweep variants when echo-to-echo evolution matters. The finite sweep results keep one `CPMGTrainResult` per parameter value and also stack echo integrals for quick heat-map style summaries.

7.3 FID

```
from spin_dynamics.parameters import set_params_ideal_fid
from spin_dynamics.workflows.fid import sim_fid_ideal

sp, pp = set_params_ideal_fid(numpts=101)
macq, fid, tvect = sim_fid_ideal(sp, pp)
```

7.4 Imaging

```
import numpy as np
from spin_dynamics.workflows import (
    make_imaging_field_maps,
    run_tuned_phase_encoded_cpmg_imaging,
    form_imaging_image,
)

rho = np.eye(8)
fields = make_imaging_field_maps(
    rho,
    b0_map=np.zeros_like(rho),
    b1_tx_map=np.ones_like(rho),
    b1_rx_map=np.ones_like(rho),
)
result = run_tuned_phase_encoded_cpmg_imaging(
    fields,
    num_echoes=4,
    ny=9,
    receive_mode="weighted",
)
image = form_imaging_image(result, mode="echo_sum")
```

Imaging workflows return raw k-space, reconstructed per-echo images, field maps, relaxation maps, gradient metadata, and sequence timing. Image formation is intentionally separated from acquisition so selected-echo, echo-summed, fitted-ρ, and fitted-T_2 displays can be compared. `make_imaging_field_maps` accepts spatial B0, transmit-B1, and receive-B1 maps, and the ideal imaging path can include an inversion-recovery preparation through `run_t1_encoded_phase_encoded_cpmg_imaging`. When `noise` is supplied, clean k-space and image fields remain in the result while noisy counterparts are stored in `kspace_noisy`, `image_noisy`, and `magnitude_noisy`.

7.5 Received-Signal Noise

Noise is opt-in and output-referred. High-level workflows accept a `NoiseSpec`, a mapping, or a scalar white-noise standard deviation:

```
from spin_dynamics.noise import NoiseSpec
from spin_dynamics.workflows import run_tuned_cpmg

noisy = run_tuned_cpmg(
    numpts=101,
    noise=NoiseSpec(model="probe", target_snr=25.0, seed=3),
)
```

`model="white"` adds flat complex receiver noise. `model="probe"` uses the probe output noise density where the workflow can supply it, which is important when the received spectrum is filtered by a tuned, untuned, or matched resonator. Result objects preserve clean deterministic fields and add noisy fields plus `NoiseMetadata` so repeated SNR studies remain reproducible.

7.6 Diffusion

```
from spin_dynamics.workflows import run_matched_diffusion_q_sweep

sweep = run_matched_diffusion_q_sweep(
    q_values=[20, 50],
    num_echoes=3,
    numpts=21,
    auto_refine_grid=True,
)
```

The compact matched-probe diffusion workflow is solver-validated through $Q = 2000$. Higher-Q cases remain a stiffness target for the current NumPy matched-probe transient solver.

7.7 Moving Isochromats

The motion layer treats inhomogeneity in the same spirit as static isochromats, but the samples move through two-dimensional B0, transmit-B1, and receive-B1 maps:

```
from spin_dynamics.motion import make_motion_field_maps_2d,
    initialize_ensemble_from_density
from spin_dynamics.sequences.motion import run_motion_cpmg_sequence

fields = make_motion_field_maps_2d([-1, 0, 1], [-1, 0, 1])
ensemble = initialize_ensemble_from_density([[0, 1, 0], [1, 2, 1], [0, 1, 0]])
motion = run_motion_cpmg_sequence(
    ensemble,
    fields,
    num_echoes=4,
    echo_spacing=1e-3,
    excitation_duration=20e-6,
    refocusing_duration=40e-6,
    gradient=(0.0, 0.1),
```

```
)
```

Lower-level helpers such as `free_precession_with_motion_step` are useful for custom trajectories. The sequence helpers split long intervals into substeps so field maps are sampled along the particle path rather than only at the beginning of an interval.

7.8 Time-Varying Fields

```
from spin_dynamics.workflows import (
    sinusoidal_field_waveform,
    run_ideal_time_varying_amplitude_sweep,
)

waveform = sinusoidal_field_waveform(num_echoes=16)
sweep = run_ideal_time_varying_amplitude_sweep(
    amplitudes=[0, 0.5, 1.0, 2.0],
    waveform=waveform,
    numpts=101,
    auto_refine_grid=True,
)
```

7.9 WURST

```
from spin_dynamics.workflows import (
    run_ideal_wurst_inversion,
    run_matched_wurst_cpmg,
)

ideal = run_ideal_wurst_inversion(numpts=101)
matched = run_matched_wurst_cpmg(num_echoes=2, numpts=51, num_steps=64)
```

7.10 Radiation Damping

```
from spin_dynamics.workflows import run_radiation_damping_fid

fid = run_radiation_damping_fid(
    probe="matched",
    fill_factor=0.7,
    equilibrium_magnetization=0.8,
    flip_angle=1.0,
    model="circuit",
    detuning=2.0e4,
)
```

Finite tuned and matched CPMG trains accept an opt-in `radiation_damping` mapping:

```
from spin_dynamics.workflows import run_tuned_cpmg_train
```

```
train = run_tuned_cpmg_train(
    numpts=51,
    num_echoes=8,
    radiation_damping={
        "fill_factor": 0.7,
        "equilibrium_magnetization": 0.8,
        "model": "circuit",
        "detuning": 2.0e4,
        "apply_during_pulses": True,
    },
)
```

The CPMG train path is deterministic and keeps the radiation-damping feedback inside the finite acquisition model. For strongly inverted samples, `simulate_nmr_maser` uses the same feedback machinery with a pumped longitudinal magnetization; it is intended as an idealized threshold and saturation diagnostic rather than a full spectrometer-control model.

7.11 Inverse Laplace Analysis

```
import numpy as np
from spin_dynamics.analysis import invert_t2, t2_kernel

echo_times = np.linspace(0.5e-3, 90e-3, 40)
t2_axis = np.logspace(-4, -1, 60)
signal = t2_kernel(echo_times, t2_axis) @ np.exp(-((np.log(t2_axis) + 4.5) ** 2))

result = invert_t2(
    signal,
    echo_times,
    t2_axis,
    regularization=5e-4,
    regularization_order=2,
)
```

For two-dimensional distributions, the separable forward model is

$$D = K_1 F K_2^{\mathsf{T}},$$

where D is the measured data, F is the unknown distribution, and K_1, K_2 are relaxation or diffusion kernels. Use `invert_t1_t2` or `invert_d_t2` for common NMR cases.

7.12 Optimization Workflows

The optimization package provides compact, plotting-free drivers for phase-program searches and result handoff:

```
from spin_dynamics.optimization import run_tuned_refocusing_multistart

opt = run_tuned_refocusing_multistart(
    num_segments=3,
    num_starts=4,
```

```
    numpts=21,
    max_passes=4,
)
```

Use the optimization examples as diagnostics rather than turnkey production pulse design: they expose refocusing, target-excitation, and inverse-excitation objective behavior while the inverse-cancellation workflow remains under validation.

8 Examples

The example scripts can be run from `PythonSpinDynamics`; each script also works from inside `examples/` by adding the local source tree to `sys.path` when needed.

```
python examples\ideal_cpmg.py --numpts 101
python examples\ideal_fid.py --numpts 101
python examples\ideal_cpmg_train.py --numpts 101 --num-echoes 8
python examples\ideal_time_varying_cpmg.py --numpts 101 --num-echoes 16
python examples\compare_cpmg_fid.py --numpts 101
python examples\export_validation_arrays.py results\validation_arrays.npz --numpts 101
python examples\tuned_probe_cpmg.py --numpts 101
python examples\probe_cpmg_compare.py --numpts 101
python examples\probe_parameter_sweeps.py --numpts 101
python examples\matched_cpmg_ir_train.py --numpts 21 --num-echoes 4 --num-tau 4
python examples\finite_probe_train_sweeps.py --numpts 21 --num-echoes 3
python examples\matched_diffusion_cpmg.py --numpts 21 --num-echoes 3
python examples\received_signal_noise.py --numpts 51
python examples\radiation_damping_fid.py --probe matched --points 401
python examples\radiation_damping_cpmg_train.py --probe tuned --numpts 21 --num-echoes 4
python examples\nmr_maser.py
python examples\heteronuclear_j_editing.py --points 33
python examples\coupled_isochromat_fields.py --points 21
python examples\diagnose_optimization_backends.py --backend all --numpts 21 --segments 3
```

Plotting examples require Matplotlib:

```
python examples\plot_ideal_workflows.py --numpts 201 --output results\ideal_workflows.png
python examples\plot_ideal_imaging.py --pixels 6 --ny 7 --output results\ideal_imaging.png
python examples\plot_custom_imaging_fields.py --pixels 8 --ny 7 --output results\
    custom_imaging_fields.png
python examples\plot_inverse_laplace.py --output results\inverse_laplace.png
python examples\plot_probe_parameter_sweep.py --probe tuned --sweep q --output results\
    tuned_q_sweep.png
python examples\plot_probe_cpmg.py --numpts 101 --output results\probe_cpmg.png
python examples\plot_finite_train_workflows.py --numpts 17 --num-echoes 4 --output results
    \finite_trains.png
python examples\plot_diffusion_sweep.py --numpts 17 --num-echoes 3 --output results\
    diffusion_sweep.png
python examples\plot_time_varying_sweep.py --numpts 51 --num-echoes 12 --output results\
    time_varying_sweep.png
python examples\plot_motion_linear.py --output results\motion_linear.png
python examples\plot_motion_diffusion_cpmg.py --output results\motion_diffusion_cpmg.png
python examples\plot_radiation_damping.py --output results\radiation_damping.png
python examples\plot_radiation_damping_detuning.py --output results\rd_detuning.png
python examples\plot_radiation_damping_cpmg_train.py --output results\rd_cpmg.png
python examples\plot_nmr_maser.py --output results\nmr_maser.png
python examples\plot_wurst_flow.py --numpts 41 --num-steps 64 --output results\wurst_flow.
    png
python examples\plot_j_editing_spectrum.py --output results\j_editing_spectrum.png
python examples\plot_j_editing_field_spread.py --output results\j_editing_field_spread.png
python examples\plot_tango_filter.py --target 160 --output results\tango_filter.png
python examples\plot_slic_two_spin.py --j-hz 7 --delta-hz 0.7 --output results\
    slic_two_spin.png
```

```
python examples\plot_optimization_workflows.py --numpts 11 --segments 2 --starts 2 --
    inverse-starts 4 --output results\optimization_workflows.png
python examples\plot_optimization_pipeline.py --numpts 11 --refocusing-segments 2 --
    refocusing-starts 2 --excitation-segments 2 --excitation-starts 2 --inverse-starts 3 --
    output results\optimization_pipeline.png
python examples\plot_nqr_powder_nutation.py --output results\nqr_powder_nutation.png
python examples\plot_nqr_population_transfer.py --output results\nqr_population_transfer.
    png
python examples\plot_nqr_slse_offset.py --output results\nqr_slse_offset.png
python examples\plot_nqr_slse_spacing.py --output results\nqr_slse_spacing.png
python examples\plot_nqr_efg_broadening.py --output results\nqr_efg_broadening.png
python examples\plot_nqr_temperature_broadening.py --output results\
    nqr_temperature_broadening.png
python examples\plot_nqr_slse_efg_broadening.py --output results\nqr_slse_efg_broadening.
    png
python examples\plot_nqr_weak_b0_spectrum.py --output results\nqr_weak_b0_spectrum.png
```

9 Validation

Validation fixtures live in `validation/fixtures`. The tests compare the Python implementation against compact MATLAB or Octave arrays for core rotations, echo conversion, FID, the `arb1D` kernel, finite acquisition, probe response, imaging, pulse-shape helpers, optimization result handling, and public workflow result shapes.

Regenerate dependency-light fixtures with Octave or MATLAB from the `PythonSpinDynamics` directory:

```
matlab -batch "run('validation/octave/generate_basic_fixtures.m')"
matlab -batch "run('validation/octave/generate_imaging_fixtures.m')"
matlab -batch "run('validation/octave/generate_optimization_fixtures.m')"
matlab -batch "run('validation/octave/generate_pulse_fixtures.m')"
```

Matched-probe fixtures require MATLAB toolboxes used by the reference implementation. Octave fixture generation skips matched-probe files when `fmincon` is unavailable.

10 API Reference

This chapter documents the intended public surface. Lower-level module imports remain available for validation and porting. For application code, start with:

```
spin_dynamics.workflows
spin_dynamics.parameters
spin_dynamics.analysis
spin_dynamics.coupling
spin_dynamics.nqr
spin_dynamics.noise
spin_dynamics.motion
spin_dynamics.sequences.motion
spin_dynamics.optimization
```

Drop into lower-level modules only when a building block is needed directly. Use `spin_dynamics.coupling` for the scoped dense scalar-coupled spin-1/2 tools described in Chapter 5. Use `spin_dynamics.nqr` for the selective pulsed NQR tools described in Chapter 6.

10.1 Workflow Results

Class	Purpose
`CPMGResult`	Asymptotic CPMG spectrum, echo, time vector, SNR, and probe label.
`CPMGTrainResult`	Finite echo-train spectra, echoes, echo integrals, and sequence timing.
`CPMGIRTrainResult`	Inversion-recovery finite trains over a `tauvect`.
`MatchedCPMGIRTrainResult`	Matched-probe specialization of the CPMG-IR train result.
`CPMGParameterSweepResult`	Asymptotic Q or mistuning sweeps.
`CPMGFiniteParameterSweepResult`	Finite-train Q or mistuning sweeps.
`ZMagnetizationSweepResult`	Matched-probe z-magnetization Q sweeps.
`IdealTimeVaryingCPMGResult`	Final echo for ideal time-varying-field CPMG.
`ProbeTimeVaryingCPMGResult`	Probe-aware time-varying-field CPMG result.
`IdealCPMGImagingResult`	Ideal phase-encoded CPMG imaging result.
`ProbeCPMGImagingResult`	Tuned or matched phase-encoded CPMG imaging result.
`ImagingEchoFitResult`	Voxel-wise mono-exponential echo-stack fit.
`ImagingNoiseStatistics`	Repeated-trial image-domain noise summary.
`MotionSequenceResult`	Moving-isochromat sequence samples and final particle ensemble.
`NoiseMetadata`	Generated received-noise parameters and realized SNR.
`MatchedDiffusionCPMGResult`	Matched diffusion-aware CPMG train.
`MatchedDiffusionQSweepResult`	Matched diffusion Q sweep.
`WURSTInversionResult`	WURST inversion profile and pulse metadata.

`MatchedWURSTCPMGResult`	Matched WURST-CPMG echoes and spectra.
`RadiationDampingFIDResult`	Radiation-damping FID simulation and analytic comparison fields.
`MultiStartOptimizationResult`	Repeated random-start pulse-optimization result.
`RefocusingOptimizationResult`	Bounded fixed-amplitude refocusing phase optimization.
`ExcitationOptimizationResult`	Bounded fixed-amplitude target-excitation or inverse-excitation optimization.
`TunedExcitationInversePipelineResult`	Selected-refocusing to excitation/inverse-excitation pipeline result.
`SLSEResult`	NQR SLSE echo train, local orientation signals, and transition metadata.
`SLSESweepResult`	NQR SLSE selected-echo amplitudes and effective decay estimates across a sweep.
`NQRFIDDistributionResult`	EFG-distribution FID, FFT spectrum, and isochromat metadata.
`SLSEAcquisitionSpectrumResult`	Finite-window SLSE echo acquisition and spectrum, including optional noise and deconvolution metadata.
`WeakB0SpectrumResult`	Static weak-B0 transition spectrum and perturbation-ratio metadata.
`PopulationTransferResult`	NQR perturbation-plus-SLSE signal, reference signal, and normalized difference.

10.2 High-Level Workflows

```
from spin_dynamics.workflows import (
    run_ideal_cpmg, run_tuned_cpmg, run_untuned_cpmg, run_matched_cpmg,
    run_ideal_cpmg_train, run_tuned_cpmg_train,
    run_untuned_cpmg_train, run_matched_cpmg_train,
    run_ideal_cpmg_ir_train, run_tuned_cpmg_ir_train,
    run_untuned_cpmg_ir_train, run_matched_cpmg_ir_train,
    run_tuned_q_sweep, run_matched_q_sweep,
    run_tuned_mistuning_sweep, run_matched_mistuning_sweep,
    run_tuned_finite_q_sweep, run_untuned_finite_q_sweep,
    run_matched_finite_q_sweep,
    run_tuned_finite_mistuning_sweep,
    run_untuned_finite_mistuning_sweep,
    run_matched_finite_mistuning_sweep,
    run_matched_z_magnetization_q_sweep,
    run_matched_diffusion_cpmg, run_matched_diffusion_q_sweep,
    run_ideal_time_varying_cpmg_final,
    run_tuned_time_varying_cpmg_final,
    run_untuned_time_varying_cpmg_final,
    run_matched_time_varying_cpmg_final,
    run_ideal_time_varying_amplitude_sweep,
    run_tuned_time_varying_amplitude_sweep,
    run_untuned_time_varying_amplitude_sweep,
    run_matched_time_varying_amplitude_sweep,
    sinusoidal_field_waveform,
    run_ideal_wurst_inversion,
```

```
    run_matched_wurst_inversion,
    run_matched_wurst_cpmg,
    run_radiation_damping_fid,
)
```

The main CPMG runners share `numpts` and `maxoffs`. Finite train runners add `num_echoes`, relaxation constants, rephasing controls, optional `auto_refine_grid`, and optional worker controls. Probe-aware train runners also accept `q_value` and `mistuning_offset` where the underlying probe model supports them.

10.3 Imaging API

```
from spin_dynamics.workflows import (
    ImagingFieldMaps,
    make_imaging_field_maps,
    load_imaging_field_maps_npz,
    run_ideal_phase_encoded_cpmg_imaging,
    run_tuned_phase_encoded_cpmg_imaging,
    run_matched_phase_encoded_cpmg_imaging,
    run_t1_encoded_phase_encoded_cpmg_imaging,
    reconstruct_image_from_kspace,
    form_imaging_image,
    fit_imaging_echo_decay,
    summarize_imaging_noise_trials,
)
```

Prefer the `phase_encoded` names for new code. The shorter compatibility aliases remain available:

```
run_ideal_cpmg_imaging
run_tuned_cpmg_imaging
run_matched_cpmg_imaging
```

10.4 Parameter Constructors

```
from spin_dynamics.parameters import (
    set_params_ideal,
    set_params_ideal_fid,
    set_params_tuned_orig,
    set_params_untuned_orig,
    set_params_matched_orig,
    set_params_tuned_spa,
    set_params_untuned_spa,
    set_params_matched_spa,
    set_params_tuned_jmr,
    set_params_untuned_jmr,
    set_params_matched_jmr,
)
```

These constructors return dataclass instances mirroring MATLAB `sp`, `pp`, and `params` structures. They accept optional `numpts` arguments for lightweight tests and examples while preserving MATLAB defaults when omitted.

10.5 Analysis API

```
from spin_dynamics.analysis import (
    t2_kernel, t1_kernel, diffusion_kernel, laplace_kernel,
    invert_laplace_1d, invert_laplace_2d,
    invert_t2, invert_t1, invert_t1_t2, invert_d_t2,
    default_regularization_strengths,
    estimate_noise_rms_from_snr,
    expected_residual_norm_from_snr,
    select_regularization_1d,
    select_regularization_2d,
)
```

For one-dimensional relaxation distributions, use:

```
invert_t2
invert_t1
```

For separable two-dimensional maps, use:

```
invert_t1_t2
invert_d_t2
```

The `select_regularization_*` helpers choose Tikhonov strengths from an RMS SNR estimate.

10.6 Optimization Diagnostics

Optimization helpers expose compact NumPy/SciPy-compatible diagnostics for phase-program searches:

```
from spin_dynamics.optimization import run_tuned_refocusing_multistart

result = run_tuned_refocusing_multistart(num_segments=3, num_starts=4)
```

The plotting-free analysis helpers convert MATLAB-style result cells into score summaries, and the tuned excitation/inverse pipeline carries a selected refocusing candidate into bounded target-excitation and inverse-cancellation tests. The inverse stage is still best treated as a parity and diagnostics surface while strong inverse-pulse cancellation remains under validation.

10.7 Core Numerical API

```
from spin_dynamics.core.echo import (
    calc_time_domain_echo,
    calc_time_domain_echo_arb,
    calc_fid_time_domain,
)
from spin_dynamics.core.rotations import (
    calc_rotation_matrix,
    calc_rot_axis_arba3,
    calc_rot_axis_arba4,
    calc_v0crit,
    sim_spin_dynamics_asymp_mag3,
```

```
    sim_spin_dynamics_exc,
)
from spin_dynamics.core.kernels import (
    sim_spin_dynamics_arb7,
    sim_spin_dynamics_arb10,
    sim_spin_dynamics_arb10_chunked,
    sim_spin_dynamics_arb10_diffusion,
    sim_spin_dynamics_arb10_radiation_damping,
)
from spin_dynamics.core.isochromats import (
    analyze_rephasing,
    check_rephasing,
    estimate_rephase_time,
    recommended_numpts_for_rephasing,
)
```

Core functions are intended for validation, debugging, and continued porting. They are closer to MATLAB's array contracts than the workflow layer.

10.8 Probe and Pulse API

```
from spin_dynamics.probes.tuned import (
    tuned_probe_lp,
    tuned_probe_rx,
    tuned_probe_rx_tf,
    calc_rot_axis_tuned_probe_lp_orig2,
    calc_masy_tuned_probe_lp_orig,
)
from spin_dynamics.probes.untuned import (
    untuned_probe_lp,
    untuned_probe_rx,
    untuned_probe_rx_tf,
    calc_rot_axis_untuned_probe_lp,
    calc_masy_untuned_probe_lp,
)
from spin_dynamics.probes.matched import (
    matching_network_design2,
    find_coil_current,
    find_coil_current_wurst,
    matched_probe_rx,
    calc_rot_axis_matched_probe,
    calc_masy_matched_probe_orig,
    calc_masy_matched_nut,
)
from spin_dynamics.pulses import (
    quantize_phase,
    create_wurst_pulse,
    adjust_untuned_segment_lengths,
    tuned_rectangular_pulse_response,
    untuned_rectangular_pulse_response,
    matched_rectangular_pulse_response,
    matched_wurst_pulse_response,
)
```

10.9 Noise, Motion, and Radiation Damping

```
from spin_dynamics.noise import (
    NoiseSpec,
    NoiseMetadata,
    MatchedFilterSNRResult,
    add_received_noise,
    estimate_matched_filter_snr,
    ideal_noise_density,
    tuned_probe_output_noise_density,
    untuned_probe_output_noise_density,
    matched_probe_output_noise_density,
)
from spin_dynamics.motion import (
    ParticleEnsemble,
    MotionFieldMaps2D,
    make_motion_field_maps_2d,
    initialize_ensemble_from_density,
    advect_diffuse_positions,
    move_ensemble,
    apply_free_precession,
    apply_rf_rotation,
    free_precession_with_motion_step,
    receive_signal,
    transverse_b1_magnitude,
)
from spin_dynamics.sequences.motion import (
    MotionSequenceStep,
    MotionSequenceResult,
    make_motion_cpmg_sequence,
    run_motion_sequence,
    run_motion_cpmg_sequence,
)
from spin_dynamics.radiation_damping import (
    radiation_damping_time,
    proton_thermal_magnetization_density,
    water_proton_sample,
    hyperpolarized_proton_sample,
    normalized_radiation_damping_weights,
    radiation_damping_probe_from_tuned,
    radiation_damping_probe_from_matched,
    initial_state_from_flip_angle,
    analytic_radiation_damping_envelope,
    simulate_radiation_damping,
    simulate_radiation_damping_fid,
    simulate_nmr_maser,
)
```

10.10 Scalar Coupling API

```
from spin_dynamics.coupling import (
    CoupledSpinSystem,
```

```
    CoupledIsochromatEnsemble,
    CoupledIsochromatStep,
    CoupledIsochromatSequenceResult,
    coupled_spin_system,
    coupled_isochromat_ensemble,
    free_precession_step,
    rf_step,
    simulate_coupled_isochromat_sequence,
    spin_operator,
    total_operator,
    product_operator,
    zeeman_hamiltonian,
    secular_j_hamiltonian,
    isotropic_j_hamiltonian,
    rf_hamiltonian,
    equilibrium_density,
    evolve_density,
    expectation,
    j_modulation_curve,
    proton_detected_j_modulation,
    carbon_detected_j_modulation,
    tango_b_filter,
    fit_known_j_spectrum,
    JEditingFitResult,
    simulate_slic_spectrum,
    two_spin_slic_transfer_time,
    SLICSpectrumResult,
)
```

Object	Purpose
`CoupledSpinSystem`	Validated small spin-1/2 system with offsets and scalar couplings in hertz.
`CoupledIsochromatEnsemble`	Ensemble of local B0/B1 field samples for one coupled-spin system.
`free_precession_step`, `rf_step`	Sequence-step builders with optional time-varying B0/B1 overrides.
`simulate_coupled_isochromat_sequence`	Dense coupled-spin sequence propagation over an isochromat ensemble.
`spin_operator`, `total_operator`, `product_operator`	Dense spin-1/2 operator builders.
`zeeman_hamiltonian`	Rotating-frame offset Hamiltonian in radians per second.
`secular_j_hamiltonian`	Weak-coupling $I_z I_z$ scalar Hamiltonian.
`isotropic_j_hamiltonian`	Isotropic scalar Hamiltonian for strongly coupled spin-1/2 systems.
`rf_hamiltonian`	RF nutation Hamiltonian for selected spins.
`equilibrium_density`, `evolve_density`, `expectation`	Minimal dense density-matrix utilities.
`j_modulation_curve`	Sum of cosine J-modulation components.
`proton_detected_j_modulation`	Proton-detected low-field J-editing response.

`carbon_detected_j_modulation`	Carbon-detected $\cos^n$ J-editing response for CH_n groups.
`tango_b_filter`	Ideal TANGO-B scalar-coupling filter envelope.
`fit_known_j_spectrum`	Least-squares amplitudes for supplied J-coupling positions.
`simulate_slic_spectrum`	Spin-lock response scan for dense coupled-spin systems.

10.11 NQR API

```
from spin_dynamics.nqr import (
    QuadrupolarSite,
    NQRTransition,
    NQREigensystem,
    spin_matrices,
    quadrupole_frequency_scale_hz,
    quadrupole_hamiltonian,
    zeeman_hamiltonian,
    nqr_hamiltonian,
    diagonalize_site,
    OrientationSample,
    single_crystal_orientation,
    powder_average_grid,
    b0_b1_powder_average_grid,
    b0_powder_average_grid,
    SelectivePulse,
    apply_selective_pulse,
    transition_drive_scale,
    NQRRelaxationModel,
    slse_sequence,
    simulate_slse,
    simulate_slse_offset_sweep,
    simulate_slse_spacing_sweep,
    gaussian_efg_distribution,
    temperature_efg_distribution,
    simulate_fid_efg_distribution,
    simulate_slse_acquisition_spectrum,
    simulate_weak_b0_spectrum,
    weak_field_ratio,
    zeeman_frequency_hz,
    simulate_population_transfer,
    SLSEResult,
    PopulationTransferResult,
    WeakB0SpectrumResult,
)
```

Object	Purpose
`QuadrupolarSite`	One quadrupolar nucleus in the EFG principal-axis system.

`spin_matrices`	Dense angular-momentum operators for arbitrary integer or half-integer spin.
`quadrupole_frequency_scale_hz`	Spin-dependent Hamiltonian scale for spin-1 and spin-3/2 frequency conventions.
`quadrupole_hamiltonian`	Zero-field quadrupole Hamiltonian in radians per second.
`zeeman_hamiltonian`	Optional weak-B0 perturbation in radians per second.
`diagonalize_site`	Energy levels, transition frequencies, and transition dipole vectors.
`OrientationSample`	One EFG orientation relative to lab RF and optional B0 fields.
`single_crystal_orientation`	One fixed orientation sample.
`powder_average_grid`	Normalized spherical powder quadrature grid.
`b0_b1_powder_average_grid`	Powder grid with correlated static-field and RF-field directions.
`SelectivePulse`	Rectangular selective pulse on one transition.
`apply_selective_pulse`	Embedded two-level RF propagation in the energy basis.
`NQRRelaxationModel`	Phenomenological Liouville-space population/coherence relaxation rates.
`slse_sequence`	Convenience builder for SLSE detection.
`simulate_slse`	Selective-pulse SLSE echo train for a fixed orientation or powder.
`simulate_slse_offset_sweep`, `simulate_slse_spacing_sweep`	SLSE diagnostics versus RF offset or pulse period.
`gaussian_efg_distribution`, `temperature_efg_distribution`	Static EFG isochromat distributions.
`simulate_fid_efg_distribution`	Time-domain FID and FFT spectrum from an EFG distribution.
`simulate_slse_acquisition_spectrum`	Finite-window acquired SLSE echo and spectrum, with noise and optional deconvolution.
`simulate_weak_b0_spectrum`	Static weak-B0 transition spectrum for spin-1 or spin-3/2.
`weak_field_ratio`, `zeeman_frequency_hz`	Weak-field regime checks based on $\lvert\gamma B_0\rvert/\nu_{\mathrm{ref}}$.
`simulate_population_transfer`	Perturbation pulse plus SLSE detection for 2D NQR diagnostics.

10.12 Optimization API

```
from spin_dynamics.optimization import (
    spa_pulse_list,
    rectangular_refocusing_lengths,
    evaluate_tuned_refocusing_pulse,
    evaluate_untuned_refocusing_pulse,
    evaluate_matched_refocusing_pulse,
    evaluate_ideal_v0crit_refocusing_pulse,
    evaluate_ideal_v0crit_excited_refocusing_pulse,
```

```
    evaluate_ideal_time_varying_refocusing_pulse,
    optimize_tuned_refocusing_phases,
    optimize_untuned_refocusing_phases,
    optimize_matched_refocusing_phases,
    optimize_ideal_v0crit_refocusing_phases,
    optimize_ideal_v0crit_excited_refocusing_phases,
    optimize_ideal_time_varying_refocusing_phases,
    run_tuned_refocusing_multistart,
    run_untuned_refocusing_multistart,
    run_matched_refocusing_multistart,
    run_ideal_v0crit_refocusing_multistart,
    run_ideal_v0crit_excited_refocusing_multistart,
    run_ideal_time_varying_refocusing_multistart,
    evaluate_tuned_excitation_pulse,
    evaluate_tuned_inverse_excitation_pulse,
    optimize_tuned_excitation_phases,
    optimize_tuned_inverse_excitation_phases,
    run_tuned_excitation_multistart,
    run_tuned_inverse_excitation_multistart,
    run_tuned_excitation_inverse_pipeline,
    multistart_to_matlab_results,
    save_multistart_results_npz,
    load_optimization_results,
    summarize_matlab_results,
    select_matlab_result_program,
    analyze_matlab_optimization_results,
    analyze_tuned_inverse_result_pair,
)
```

The optimization layer is useful for compact pulse-design studies and MATLAB-style result inspection. Broad historical `fmincon` parity and strong inverse-excitation cancellation parity remain validation targets.

11 Known Gaps

The Python port is intentionally conservative where MATLAB parity still matters. The largest remaining gaps are broader high-Q matched-probe validation, full historical MATLAB `.mat` structure parity, broad `fmincon` parity, paper-level scalar-coupling fixture comparisons, relaxation-aware coupled-spin pulse-sequence runners, full nonselective quadrupolar pulse propagation, experimental NQR fixture comparisons, and acceleration backends beyond the current NumPy implementation.

New work should add small reference fixtures first, then a NumPy implementation, then optional acceleration or plotting once numerical behavior is pinned down.